

Math 8 — Q1 Mid-Quarter Test · ANSWER KEY

Mid-Q1 Content · 15 questions

Teacher key. Student answers may use equivalent forms ($3^6 = 729$, $7^3 = 343$, $2^{12} = 4,096$). Scientific notation may be written with $\times 10^n$ or "e" notation ($5.1\text{e-}4$).

1. Convert 3.2 km to millimeters.

Answer: 3,200,000 mm

2. Convert 45 cm to meters.

Answer: 0.45 m

3. A student measured 24.5 g; actual mass is 25 g. What is the percent error?

Answer: 2%

4. Compute: $-15 + 8$.

Answer: -7

5. Compute: $(-4) \times (-7)$.

Answer: 28

6. Compute: $(-12) - (-4)$.

Answer: -8

7. A blue whale weighs 180,000 kg (actual). A scientist measured it as 175,000 kg — an error of 5,000 kg. A house cat weighs 8 kg (actual). Another scientist measured it as 7 kg — an error of 1 kg. Which measurement is MORE accurate?

Answer: whale (whale $\approx 2.78\%$ error vs. cat = 12.5% error)

Why? Percent error compares the mistake to the size of what's being measured. The whale error of 5,000 kg sounds huge, but relative to 180,000 kg it's less than 3%. The cat error of 1 kg is much smaller in absolute terms but 12.5% of 8 kg. Smaller percent error = more accurate.

8. Write 82,500,000 in scientific notation.

Answer: 8.25×10^7

9. Write 0.00051 in scientific notation.

Answer: 5.1×10^{-4}

10. Compute: $(4 \times 10^5)(3 \times 10^2)$. Give in scientific notation.

Answer: 1.2×10^8

11. Compute: $(9 \times 10^8) \div (3 \times 10^3)$. Give in scientific notation.

Answer: 3×10^5

12. Convert 3.2×10^{-2} to standard form.

Answer: 0.032

13. Simplify (Laws of Exponents): $3^4 \cdot 3^2$.

Answer: $3^6 = 729$

14. Simplify: $(7^5) \div (7^2)$.

Answer: $7^3 = 343$

15. Simplify: $(2^3)^4$.

Answer: $2^{12} = 4,096$